

HOLZDORF, GermanyWMO: **104760**Lat: **51.7658N**Lon: **13.1667E**Elev: **102**StdP: **100.11**Time Zone: **1.00 (EUC)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.5	-8.9	-14.9	1.0	-10.4	-11.6	1.4	-8.0	11.4	7.1	10.3	6.9	2.0	100	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	31.9	19.8	29.9	18.9	27.8	18.2	20.9	28.7	20.0	27.2	19.1	25.6	3.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	13.8	22.5	17.8	12.9	22.0	16.8	12.1	21.6	61.0	28.6	57.8	27.1	54.7	25.6	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 9.3	8.0	7.0		-16.5	35.3	4.0	2.2	-19.4	36.9	-21.7	38.1	-24.0	39.4	-26.9	40.9	
(5)				-16.7	22.8	4.0	1.0	-19.6	23.5	-21.9	24.1	-24.1	24.6	-27.0	25.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	0.4	1.4	4.6	9.4	13.9	17.2	19.3	18.7	14.6	9.8	4.9	1.9
	DBStd	7.79	5.28	4.57	4.02	4.05	3.86	3.55	3.35	3.17	3.30	3.70	3.86	4.41
	HDD10.0	1242	299	242	172	59	9	1	0	0	2	50	156	252
	HDD18.3	3291	557	475	427	268	145	62	28	34	119	264	402	510
	CDD10.0	1141	1	1	3	41	131	218	289	268	141	44	4	1
	CDD18.3	150	0	0	0	1	9	29	59	44	8	0	0	0
	CDH23.3	1958	0	0	0	17	159	410	706	543	119	3	0	0
	CDH26.7	666	0	0	0	2	41	139	272	189	23	0	0	0
Wind	WSAvg	3.1	3.6	3.5	3.4	3.0	3.0	3.0	3.0	2.8	2.8	3.0	3.1	3.5
Precipitation	PrecAvg	561	44	32	38	30	55	53	73	61	47	36	46	44
	PrecMax	778	74	60	98	91	116	133	185	186	143	75	122	84
	PrecMin	356	5	7	8	1	6	13	20	16	9	7	0	12
	PrecStd	103	20	15	24	20	30	28	39	37	29	19	25	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	13.9	18.9	25.5	30.3	33.1	35.0	34.2	29.2	23.0	16.0	13.0
		MCWB	9.8	8.7	11.7	15.2	18.4	20.1	20.2	20.3	18.3	16.2	12.9	10.2
	2%	DB	10.8	11.5	16.0	22.5	27.5	30.2	32.1	31.1	26.8	19.9	13.8	10.9
		MCWB	8.4	8.2	9.9	13.7	17.0	19.0	19.7	19.7	17.4	14.6	10.9	8.7
	5%	DB	9.0	9.8	13.3	20.2	24.8	28.0	29.8	28.7	24.2	17.9	12.1	9.2
		MCWB	7.1	6.8	8.5	12.6	15.7	18.2	19.1	18.7	16.7	13.8	9.7	7.4
	10%	DB	7.2	8.0	11.1	18.0	22.4	25.2	27.2	26.2	21.8	15.9	10.7	7.9
		MCWB	5.7	5.9	7.8	11.5	14.4	17.0	18.4	17.8	15.5	12.8	8.7	6.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	9.7	12.2	15.7	19.6	22.0	22.1	21.9	20.0	16.9	13.2	10.7
		MCDB	12.3	12.5	17.5	23.8	28.1	29.7	30.8	30.8	26.4	21.5	15.5	12.6
	2%	WB	8.6	8.5	10.5	14.4	18.0	20.2	20.8	20.8	18.3	15.3	11.4	8.9
		MCDB	10.2	11.1	14.4	21.1	24.8	27.1	28.6	28.5	24.1	18.9	13.3	10.5
	5%	WB	7.1	7.1	9.3	13.2	16.7	19.0	19.9	19.7	17.2	14.2	10.0	7.6
		MCDB	8.7	9.2	12.8	19.1	23.1	26.0	27.3	26.4	22.6	17.3	11.6	9.1
	10%	WB	5.7	5.7	8.0	12.0	15.3	17.7	19.0	18.7	16.3	13.0	8.9	6.2
		MCDB	7.2	7.8	10.9	17.1	20.8	24.2	25.4	24.4	21.0	15.5	10.4	7.6
Mean Daily Temperature Range	5% DB	MDBR	5.8	7.1	9.1	12.1	12.5	12.3	12.0	12.6	11.3	9.0	6.5	5.3
		MCDBR	6.7	9.6	14.0	16.2	16.8	16.9	17.0	16.8	15.6	12.2	8.0	6.3
		MCWBR	5.1	6.5	8.6	8.7	7.8	7.2	6.6	7.2	7.7	7.6	5.6	4.9
	5% WB	MCDBR	6.3	8.5	12.5	14.8	14.6	15.0	14.1	14.0	13.3	11.2	7.3	5.9
		MCWBR	5.0	6.1	8.2	8.1	7.2	6.9	5.9	6.7	7.0	7.3	5.5	4.9
Clear-Sky Solar Irradiance	taub	0.320	0.345	0.387	0.417	0.418	0.414	0.433	0.426	0.388	0.369	0.344	0.314	
	taud	2.341	2.317	2.215	2.194	2.236	2.275	2.230	2.266	2.346	2.364	2.374	2.354	
	Ebn at Noon	678	765	795	817	836	842	816	801	791	726	642	620	
	Edh at Noon	63	85	113	130	131	127	131	120	99	81	62	54	
All-Sky Solar Radiation	RadAvg	0.70	1.39	2.40	3.97	4.88	5.20	4.90	4.24	3.06	1.75	0.83	0.54	
	RadStd	0.09	0.24	0.29	0.51	0.58	0.42	0.63	0.36	0.37	0.20	0.12	0.06	